

```
name="Multiplicant_B"><br>
Result: <input type="number" id="answer"
name="Multiplicant_C"><br>
<input type="button" value="Multiplication_compute_on_C++"
onclick="Multiply()">
</form>
</body>
</html>
```

Please note that in the above HTML code, *myoperations* is a class object. And *MultOfNumbers* is its public *Q\_INVOKABLE* class method.

## How to call the C++ methods from the Web page using the Qt-WebKit framework

Let's say, I have the following class that has the *Q\_INVOKABLE* method, *MultOfNumbers*.

```
class MyJavaScriptOperations : public QObject {
    Q_OBJECT
public:
    Q_INVOKABLE qint32 MultOfNumbers(int a, int b) {
        qDebug() << a * b;
        return (a*b);
    }
};
```

This class object should be added to the JavaScript window object by the following API:

```
addToJavaScriptWindowObject("name of the object", new (class
that can be accessed))
```

Here is the entire program:

```
#include <QtGui/QApplication>
#include <QApplication>
#include <QDebug>
#include <QWebFrame>
#include <QWebPage>
#include <QWebView>

class MyJavaScriptOperations : public QObject {
    Q_OBJECT
public:
    Q_INVOKABLE qint32 MultOfNumbers(int a, int b) {
        qDebug() << a * b;
        return (a*b);
    }
};

int main(int argc, char *argv[])
{
    QApplication a(argc, argv);
```

```
QWebView *view = new QWebView();
view->resize(400, 500);
view->page()->mainFrame()->addToJavaScriptWindowObject("myope
rations", new MyJavaScriptOperations);
view->load(QUrl("./index.html"));
view->show();
return a.exec();
}
#include "main.moc"
```

The output is given in Figure 1.

## How to install a callback from C++ code to the Web page using the Qt-WebKit framework

We have already seen the call to C++ methods by JavaScript. Now, how about a callback from C++ to JavaScript? Yes, it is possible with the Qt-WebKit. There are two ways to do so. However, for the sake of neatness in design, let's discuss only the Signals and Slots mechanisms for the JavaScript callback.

## Installing Signals and Slots for the JavaScript function

Here are the steps that need to be taken for the callback to be installed:

- Add a JavaScript window object to the *JavaScriptWindowObjectCleared* slot.
- Declare a signal in the class.
- Emit the signal.
- In JavaScript, connect the signal to the JavaScript function slot.

Here is the syntax to help you connect:

```
<JavaScript_window_object>.<signal_name>.connect(<JavaScript
function name>);
```

Note, you can make a callback to JavaScript only after the Web page is loaded. This can be ensured by connecting to the Slot emitted by the Signal *loadFinished()* in the C++ application.

Let's look at a real example now. This will fire a callback once the Web page is loaded.

The callback should be addressed by the JavaScript function, which will show up an alert window.

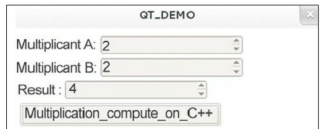


Figure 1: QT DEMO output